Nasal/sublingual/troche gray-market peptide preparations (mucosal route): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for Nasal/sublingual/troche gray-market peptide preparations (mucosal route) (Peptides, SARMs, and Research Chemicals), evaluated on the finished product as sold. Every analyte is held to the ICH Q3D(R2) elemental-impurity limit for the parenteral route (the harmonized pharmaceutical standard, enforceable via USP <232>), measured on every lot by ICP-MS. Certification is heavy-metal safety only and does not endorse the efficacy or legality of these products.

1 | How every number on this standard is set

HMTc Method v2.0 — no-regulatory-anchor (control) form. Government-limit lookup returns nothing for this class, so cells resolve to control or a substantiated not_material; ICH Q3D / USP <232> route-appropriate PDEs are toxicological anchors only, never product-content limits.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	500	human-grade read-across (ICH)

Cadmium (Cd)	200	human-grade read-across (ICH)
Total arsenic (tAs)	1,500	human-grade read-across (ICH)
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	300	human-grade read-across (ICH)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	2,000	human-grade read-across (ICH)
Tin (Sn)	60,000	human-grade read-across (ICH)
Aluminium (Al)	25	human-grade read-across (US FDA)
Total chromium (Cr)	110,000	human-grade read-across (ICH)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	ICH — human-food analog
Cadmium (Cd)	200	ICH — human-food analog
Total arsenic (tAs)	1,500	ICH — human-food analog
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	300	ICH — human-food analog
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	2,000	ICH — human-food analog
Tin (Sn)	60,000	ICH — human-food analog
Aluminium (Al)	25	US FDA — human-food analog

Total chromium (Cr)	110,000	ICH — human-food analog
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Nasal/sublingual/troche gray-market peptide preparations (mucosal route): HMTc standard* (Standard No. 10.83138/hmtc.std.26.peptides-nasal-sublingual). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.26.peptides-nasal-sublingual

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References

[1] ICH — human-food maximum levels
Binds Pb, Cd, tAs, tHg, Ni, Sn, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] US FDA — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/peptides-nasal-sublingual. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-09. The full dated record is maintained at https://heavymetalcertified.com/standards/peptides-nasal-sublingual.